YOUNGYOON CHOI

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RESEARCH INTERESTS

Generative modeling, particularly in video, grounded in theory-driven approaches. Experience in 3D and reinforcement learning, with broad interest in their integration with generative models.

PAPERS

Diverse Motion Customization via Control-based Dynamic Optimization
Youngyoon Choi*, Kihyun Kim*, Jeongwoo Shin, Joonseok Lee
NeurIPS 2026 (under review)
Keywords: Motion Customization, Stochastic Optimal Control, Text-to-video generation

Trajectory-guided Referred Motion Segmentation
Yunjeong Choi, Hyunjin Lee, **Youngyoon Choi**, Eunyi Lyou, Yoori Oh, Joonseok Lee
ECCV 2026 (under review)
Keywords: Referring Video Object Segmentation, Point Trajectory, Motion

Improving 3D MRI Generation via Latent Space Decorrelation
Wonyoung Jang*, Junho Lee*, **Youngyoon Choi**, Seunggeun Lee, Joonseok Lee
Medical Image Understanding and Analysis (MIUA) 2026
Keywords: Brain Imaging, Flow matching, Representation Learning

GaussianVideo: Efficient Video Representation and Compression by Gaussian Splatting
Inseo Lee, **Youngyoon Choi**, Joonseok Lee
CVPR Workshop on Perception Beyond the Visible Spectrum 2025
Keywords: 3D Gaussian Splatting, Video Compression, Neural Rendering

EDUCATION

M.S. in Graduate School of Data Science — Mar 2024 – Present
Seoul National University, South Korea
Advisor: Joonseok Lee · Expected Feb 2027

B.A. in Psychology (Interdisciplinary Program in Brain, Mind, and Behavio) — Mar 2019 – Feb 2024
Seoul National University, South Korea
CGPA: 3.96 / 4.3 · Degree Honors: Summa Cum Laude

RESEARCH EXPERIENCE

Graduate Researcher, Visual Information Processing (VIP) Lab — Feb 2024 – Present
Advisor: Joonseok Lee

- Conducting research on video generation and editing, with a focus on controllable synthesis grounded in theory-driven approaches.
- Primary work applies stochastic optimal control to address content leakage in motion customization; NeurIPS 2026 under review.

Undergraduate Researcher, Computational Clinical Science (CCS) Lab — Sep 2023 – Dec 2023
Advisor: Wooyoung Ahn

- Conducted research on multimodal decision-making prediction, applying probabilistic modeling to integrate heterogeneous data.
- Received 3rd place at a research symposium.

Undergraduate Researcher, Human Factors Psychology (HFP) Lab — Jan 2023 – May 2023
Advisor: Sowon Hahn

- Assisted in participant recruitment and experimental procedures for a study examining how feedback modality in human-robot interaction affects exercise behavior change.

Undergraduate Researcher, Connectome Lab Jan 2022 – Dec 2022
Advisor: Jiwook Cha

- Conducted research on youth suicide, applying machine learning to identify risk factors.
- Selected for the Undergraduate Research Funding Program.

SELECTED PROJECTS

LexiMate: An End-to-End Inclusive System for Dyslexia Support
Developer / Data Engineer / Domain Expert

- Built a product-level end-to-end system for dyslexia screening and intervention using real-time multimodal data.
- Designed and developed the data collection pipeline, enabling structured acquisition of heterogeneous multimodal inputs.

Efficient Motion Generation with Mean Flows
Team Researcher

- Developed a one-step text-conditioned 3D human motion generation framework using mean flow modeling, achieving real-time synthesis with reduced inference cost over diffusion-based methods.
- Designed the continuous-time vector field model in latent motion space as the core architectural contribution.

Advancing Image Reconstruction: Evaluating BOLD and Beta Signals in fMRI-Based Brain Decoding
Team Researcher

- Explored diffusion-based generative approaches for fMRI-to-image reconstruction, evaluating output quality via CLIP, SSIM, and semantic similarity.
- Led signal analysis and benchmarking, connecting neuroscientific domain knowledge with applied generative modeling.

Honorable Military Family Benefits Platform
Backend Developer

- Built a backend system enabling personalized benefit matching for eligible users using PostgreSQL.
- Contributed as the primary backend developer, handling database design and query optimization.

TEACHING EXPERIENCE

Basic Computing: First Adventures in Computing Feb 2024 – Present
Led weekly hands-on Python programming classes for undergraduate students

Computational Thinking: Thinking with Computers Dec 2025
Delivered a special lecture on data analysis and visualization

Machine Learning and Deep Learning I Sep 2024 – Nov 2024

SCHOLARSHIPS & AWARDS

4th stage BK21 Scholarship Sep 2024 – Dec 2025
3rd Place, Brain-Mind-Behavior Interdisciplinary Research Presentation Dec 2023
Undergraduate Research Funding Program Jul 2022 – Dec 2022
Dean's List, Academic Scholarship Dec 2020

SKILLS

Languages Korean (Native), English (Native), Spanish (Functional)
Programming Python, R, SQL, C/C++